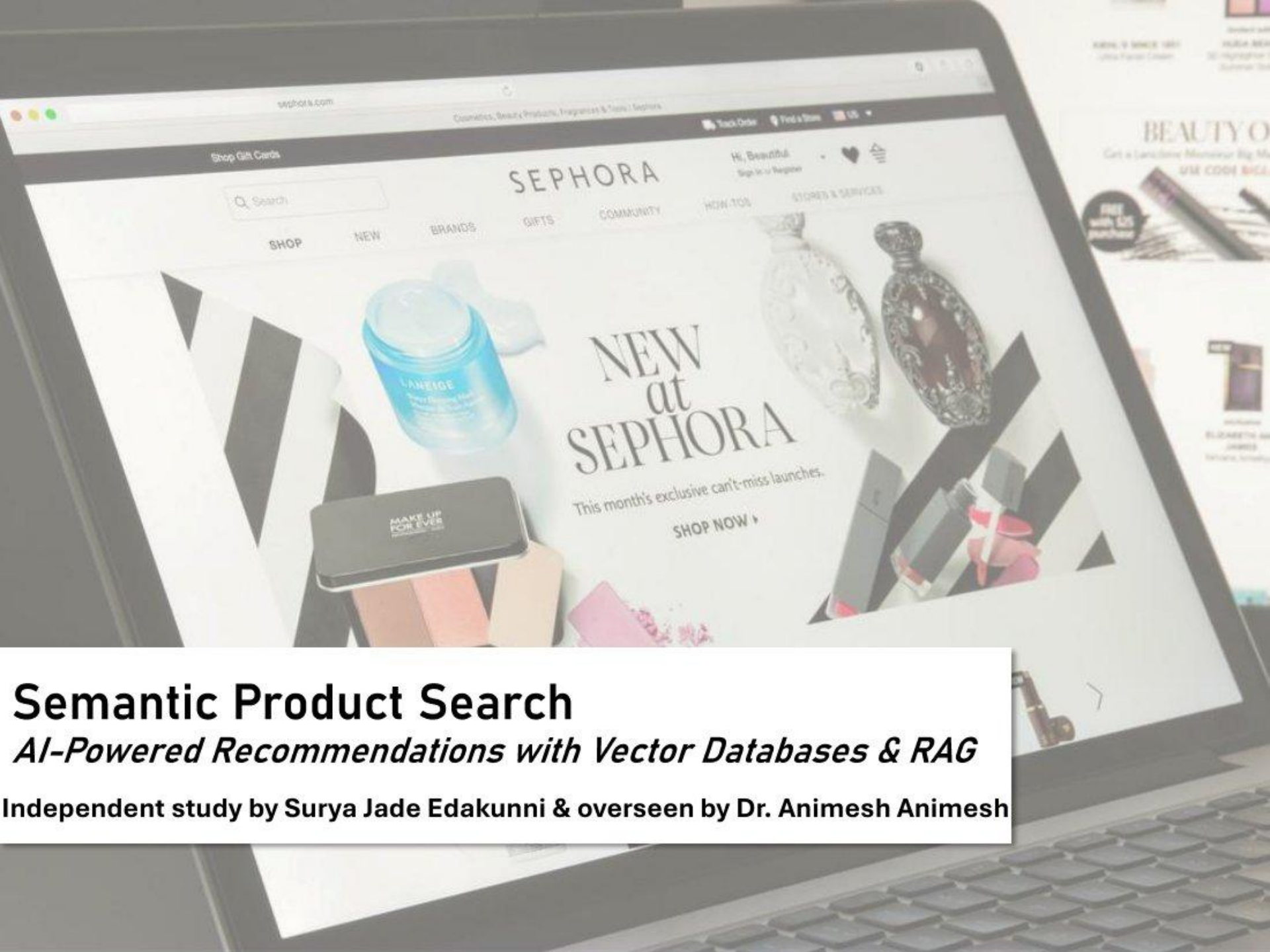


Semantic Product Search

AI-Powered Recommendations with Vector Databases & RAG

Independent study by Surya Jade Edakunni & overseen by Dr. Animesh Animesh

The Problem & Opportunity

Why Semantic Product Search?



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The Product Search Problem

Problems Identified

Keyword Search

Traditional online search often relies on exact keywords and predefined fields to retrieve products

Natural-Language Shopping

Consumers naturally describe several needs and preferences together rather than searching with isolated product terms.

The Search Gap

A relevant product may use different language from the customer's query.



"I'm looking for a moisturizer for dry, sensitive skin under $40 that won't clog my pores."



A sales associate can interpret the full request and guide the customer toward suitable products.

The same request must be interpreted by a search system that traditionally relies on keywords and predefined fields.

Research Question:

How can product search respond to what a customer means rather than only to the words they type?

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Why this Matters in Retail

The Beauty Retail Opportunity

Beauty Market Growth

$590B by 2030 is the projected global beauty market

Gen Z Spending

$9000+ annually in beauty spending

E-commerce Growth

Expected to become leading beauty purchasing channel

AI shopping Adoption

35% of E-commerce transactions could be influenced by AI

The Consumer Challenge

48% of Gen Z regret their purchases

46% of consumers trust AI assistant recommendations

Consumers face growing choice complexity while becoming increasingly receptive to AI-supported decision-making.

AI Shopping at Scale

250M+ Customers used Amazon's AI shopping assistant Rufus between 2024-2025

60%+ more likely to purchase among customers who interacted with an AI assistant during their shopping journey

Why beauty is a Strong Use Case

Large Product Catalogues

Thousands of products and variations make it difficult for consumers to identify which options best match their individual needs.

Subjective Preferences

Beauty searches often involve descriptive needs such as skin type, scent, occasion and desired product characteristics rather than simple product names.

Digital Guidance Gap

In-store shoppers can explain these preferences to a sales associate. Online, the search system must interpret the request and identify suitable products.

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From Traditional Databases to Semantic Search



Hierarchical & Network

- Relationships defined in advance
- Hierarchical systems used parent-child structures
- Difficult to adapt as data structures changed



NoSQL Databases

- Introduced more flexible record structures
- Better suited to large and varied datasets
- Improved flexibility and scale, but not semantic understanding



Vector Databases

- Store and index high-dimensional embeddings
- Compare query vectors with stored product vectors
- Retrieve products based on semantic similarity

Evolution



Relational Databases

- Organize structured data into connected tables
- SQL enables precise retrieval and manipulation
- Highly effective for prices, inventory, transactions and other structured retail data



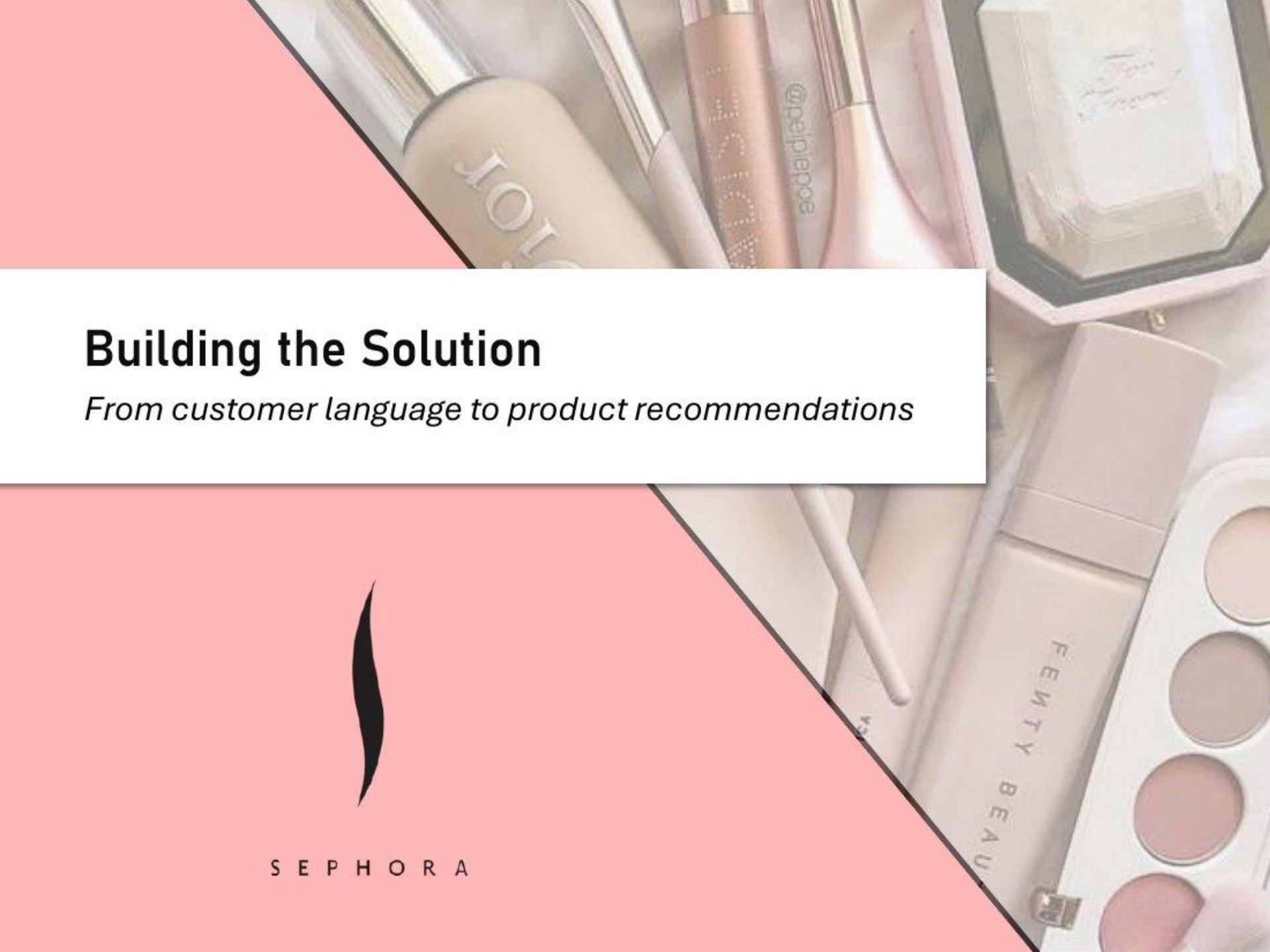
Language Embeddings

- Convert language into numerical vectors
- Similar meanings can have similar vector representations
- Enable comparison based on meaning rather than exact wording

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Building the Solution

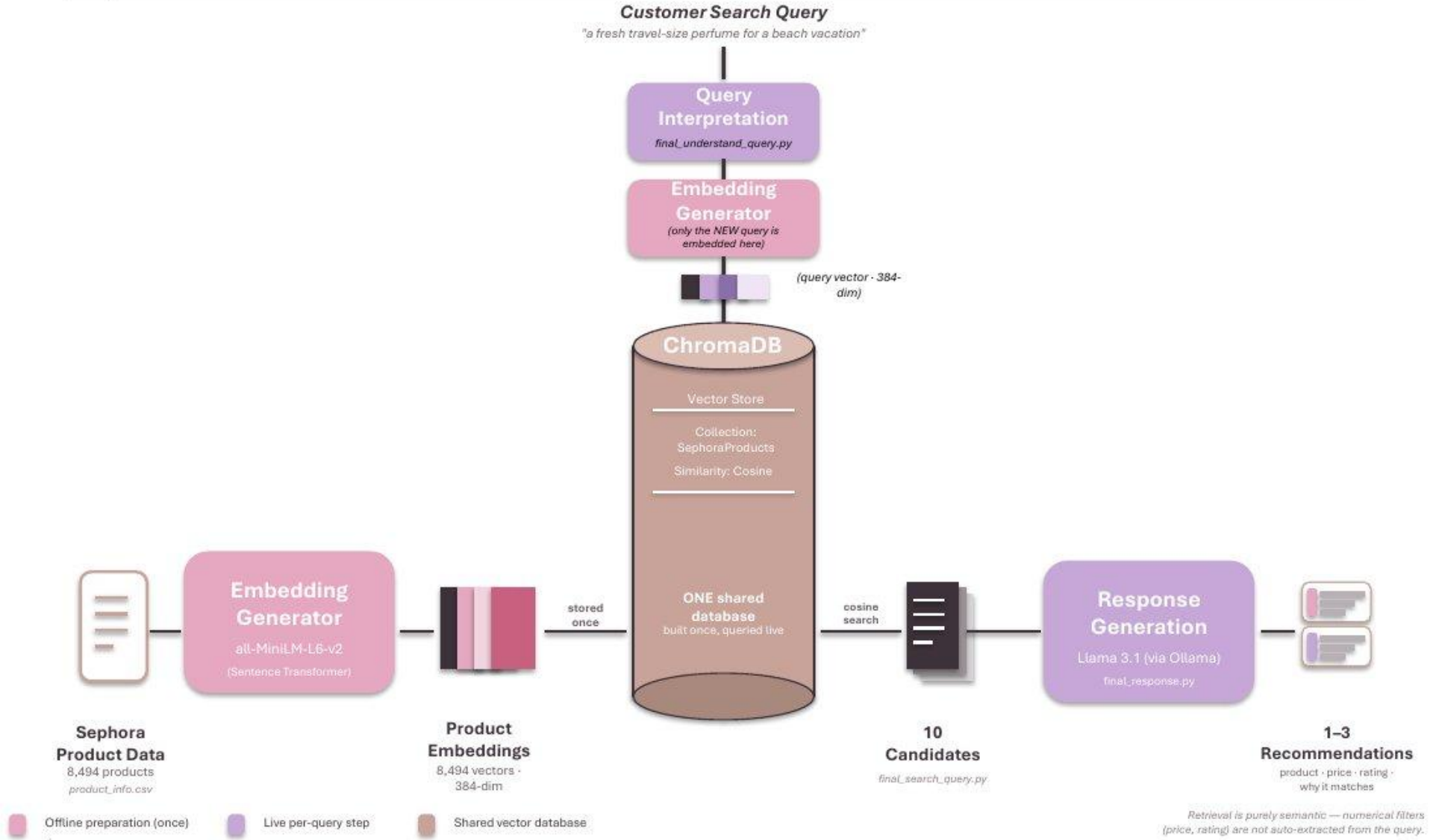
From customer language to product recommendations



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System Architecture



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From Product Data to Embeddings

Raw Product Dataset

8,494 Sephora products
across 27 features

Semantic Fields

Product name • Brand • Category • Highlights • Ingredients • Variation

Product Text

Combined into one searchable product description:

Crescent Beach Perfume Travel Spray | ABBOTT | Fragrance > Women > Rollerballs | Vegan, Fresh Scent, Cruelty-Free | 0.33 oz / 10 mL

Sentence Transformers

all-MiniLM-L6-v2

Converts each product's text into a numerical representation of its meaning

Embeddings

8,494 embeddings × 384 dimensions
[0.021, −0.184, 0.307, ..., 0.092] *(One 384-D vector per product)*

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How Vector Search Works

Text is converted into numerical vectors that represent semantic meaning. Text with similar meaning should occupy nearby regions of the vector space.

Project application

all-MiniLM-L6-v2

384 dimensions per product

Cosine Distance

Measures the orientation between two vectors rather than their absolute magnitude.

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\| A \| \| B \|}$$

$$\text{Cosine Distance} = 1 - \text{Cosine Similarity}$$

Project application

Lower distance = stronger semantic match

Embeddings

MEANING → MATCH

Metadata

Approximate Nearest Neighbor (ANN) methods retrieve likely nearest matches without exhaustively comparing every possible vector. HNSW organizes vectors as a navigable graph, allowing the search to move efficiently toward nearby candidates.

Project application

HNSW vector index

HNSW/ANN

Structured metadata answers exact conditions that similarity cannot determine.

Project Application:

price_usd • rating • reviews

Example:

price_usd ≤ 50

can be enforced as a filter rather than asking semantic similarity what "under $50" means.

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Where Semantic Search Works

EXPLICIT SEMANTIC ATTRIBUTES

QUERY

"vegan perfume for women"

TOP RESULT

Commodity Book Personal

WHY IT WORKED

Both **"vegan"** and the **women's perfume category** were explicitly represented in the product data, allowing the embedding to place the product close to the query.

DESCRIPTIVE LANGUAGE

QUERY

"woody earthy scented perfume"

TOP RESULT

REPLICA Autumn Vibes

WHY IT WORKED

The product description explicitly included **"Woody & Earthy Scent,"** creating strong semantic alignment between the query and product.

MULTIPLE PREFERENCES

QUERY

"I want a fresh travel-size perfume for women that would be good for a beach vacation"

TOP RESULT

Crescent Beach Perfume Travel Spray

WHY IT WORKED

The query combined **fresh + travel-size + women's + beach vacation**, and the system connected these ideas in one search.

Semantic search performs best when descriptive customer preferences are represented in the product data, allowing multiple needs to be connected within a single natural-language query.

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Where Semantic Search Struggles

SUBJECTIVE CONSTRAINTS

QUERY

"I want something cheap that smells expensive"

RESULT

Returned fragrance products, **but also a deodorant and a candle.**

WHY IT STRUGGLED

Semantic similarity does not enforce an exact product category, and **"cheap" has no fixed numerical meaning** unless price is applied separately as a hard filter.

MISSING PRODUCT ATTRIBUTES

QUERY

"I need an elegant perfume for my girlfriend's birthday that isn't too strong"

RESULT

The returned products were generally relevant, **but scent intensity could not be verified.**

WHY IT STRUGGLED

"Isn't too strong" cannot be reliably evaluated because scent intensity is **not represented as a field in the product data.**

SAFETY/SUITABILITY CLAIMS

QUERY

"I need a product that won't irritate my sensitive skin"

RESULT

Returned semantically relevant products, including **REN Evercalm**, but the retrieved data did not establish that the product would be non-irritating for every user.

WHY IT STRUGGLED

Semantic relevance **does not guarantee suitability or safety**. The system should not make these claims unless the underlying product data explicitly supports them.

Semantic similarity alone cannot guarantee that subjective, numerical, safety-related or unrepresented requirements in a customer's request have actually been satisfied.

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LLM Integration : Ollama + Llama 3.1

What role does the LLM play?

Vector search retrieves products based on semantic similarity; Llama 3.1 supports the pipeline before and after retrieval by interpreting the customer's request and transforming retrieved results into a readable recommendation.

OLLAMA ROLE BEFORE RETRIEVAL

Query Interpretation

INPUT
"I want a fresh travel-size perfume for women that would be good for a beach vacation."

↓ **Llama 3.1**

OUTPUT
"fresh travel-size perfume women beach vacation"

Removes conversational wording while preserving explicitly stated customer preferences.

Semantic Retrieval



ChromaDB

OLLAMA ROLE AFTER RETRIEVAL

Response Generation

INPUT
10 products retrieved by ChromaDB

↓ **Llama 3.1**

OUTPUT
1–3 Product Recommendations
Product • Brand • Price • Rating • Reviews • Category • Highlights • Size • Why it matches

Converts retrieved product information into concise, customer-facing recommendations.

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Final Output

```
Enter your query (or type 'exit' to quit): I'm looking for a warm, cozy perfume that would be good to wear in the evening during winter.

FINAL RESPONSE:
### REPLICA' By the Fireplace
**Brand:** Maison Margiela
**Price:** $160.0
**Rating:** 4.3562
**Reviews:** 2016.0
**Category:** Fragrance > Women > Perfume
**Highlights:** ['Unisex/ Genderless Scent', 'Warm &Spicy Scent']
**Size:** 3.4 oz/ 100 mL
Why it matches: This product has a warm and spicy scent, which matches the user's request for a cozy perfume for winter evenings.

### Velvet Expressive
**Brand:** Commodity
**Price:** $135.0
**Rating:** 4.4359
**Reviews:** 78.0
**Category:** Fragrance > Women > Perfume
**Highlights:** ['Vegan', 'Unisex/ Genderless Scent', 'Clean at Sephora', 'Cruelty-Free']
**Size:** 3.4 oz / 100 mL perfume spray
Why it matches: Although it doesn't explicitly mention being a warm or cozy scent, it has a unisex/genderless scent which might fit the user's request for a perfume to wear in the evening.

### REPLICA' Autumn Vibes
**Brand:** Maison Margiela
**Price:** $160.0
**Rating:** 3.7399
**Reviews:** 569.0
**Category:** Fragrance > Women > Perfume
**Highlights:** ['Woody & Earthy Scent', 'Unisex/ Genderless Scent']
**Size:** 3.4 oz/ 100 mL Eau de Toilette Spray
Why it matches: This product has a woody and earthy scent, which might not be what the user is looking for in a warm and cozy perfume, but it's the strongest match to the "warm" requirement.

Enter your query (or type 'exit' to quit):
```

Key Takeaway

The final prototype translates a **conversational request** into **ranked product recommendations** and explains the **strength of each match** using the retrieved product information.

What I learned and Where it Goes Next

Technology choices, limitations, and future applications



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Choosing the Right Database

"moisturizer for dry, sensitive skin"

Relational

How it handles the query
The request must be translated into structured fields and exact conditions before the database can retrieve matching records.
Best used when:
Data is highly structured and queries depend on precise attributes, transactions or structured relationships.

MongoDB

How it handles the query
The request can be matched against flexible product documents and their stored attributes, even when product records have different structures
Best used when:
Data is varied or semi-structured and flexibility in how records are stored is important.

ChromaDB

How it handles the query
The query is converted into an embedding and compared with stored product embeddings to retrieve products that are similar in meaning.
Best used when:
Search depends on understanding natural language and semantic similarity, as in this project.

Neo4j

How it handles the query
The search follows explicit relationships between connected entities represented as nodes and relationships.
Best used when:
The problem depends primarily on understanding and traversing relationships between entities.

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Lessons Learned



LLMs require guardrails

Testing showed that Llama 3.1 could infer numerical thresholds that customers had not explicitly stated.

LESSON : Restrict the LLM to tasks where interpretation is useful and avoid allowing it to invent hard constraints.



Semantic search cannot replace structured filtering

Concepts such as **“cheap”** do not correspond to a fixed numerical value, while explicit requirements such as price are better handled numerically.

LESSON : Semantic retrieval and structured metadata filtering solve different problems and are strongest when used together.



Retrieval is only as good as the underlying data

The system could not verify requirements such as **scent intensity** because these attributes were not represented in the product dataset.

LESSON : Better embeddings cannot recover information that does not exist in the underlying product data.



Similarity does not equal suitability

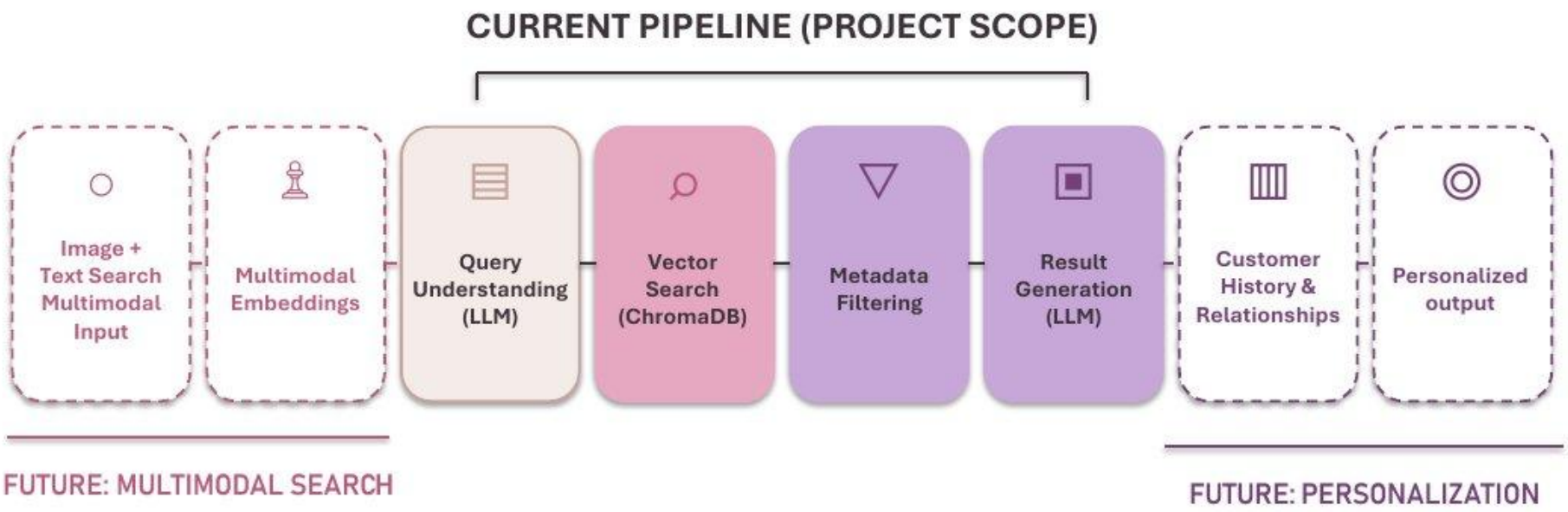
A product can be semantically close to a query without proving that every customer requirement has been satisfied, particularly for **safety or suitability claims**.

LESSON : Recommendations should remain grounded in explicit product evidence rather than treating semantic similarity as verification.

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Future Directions

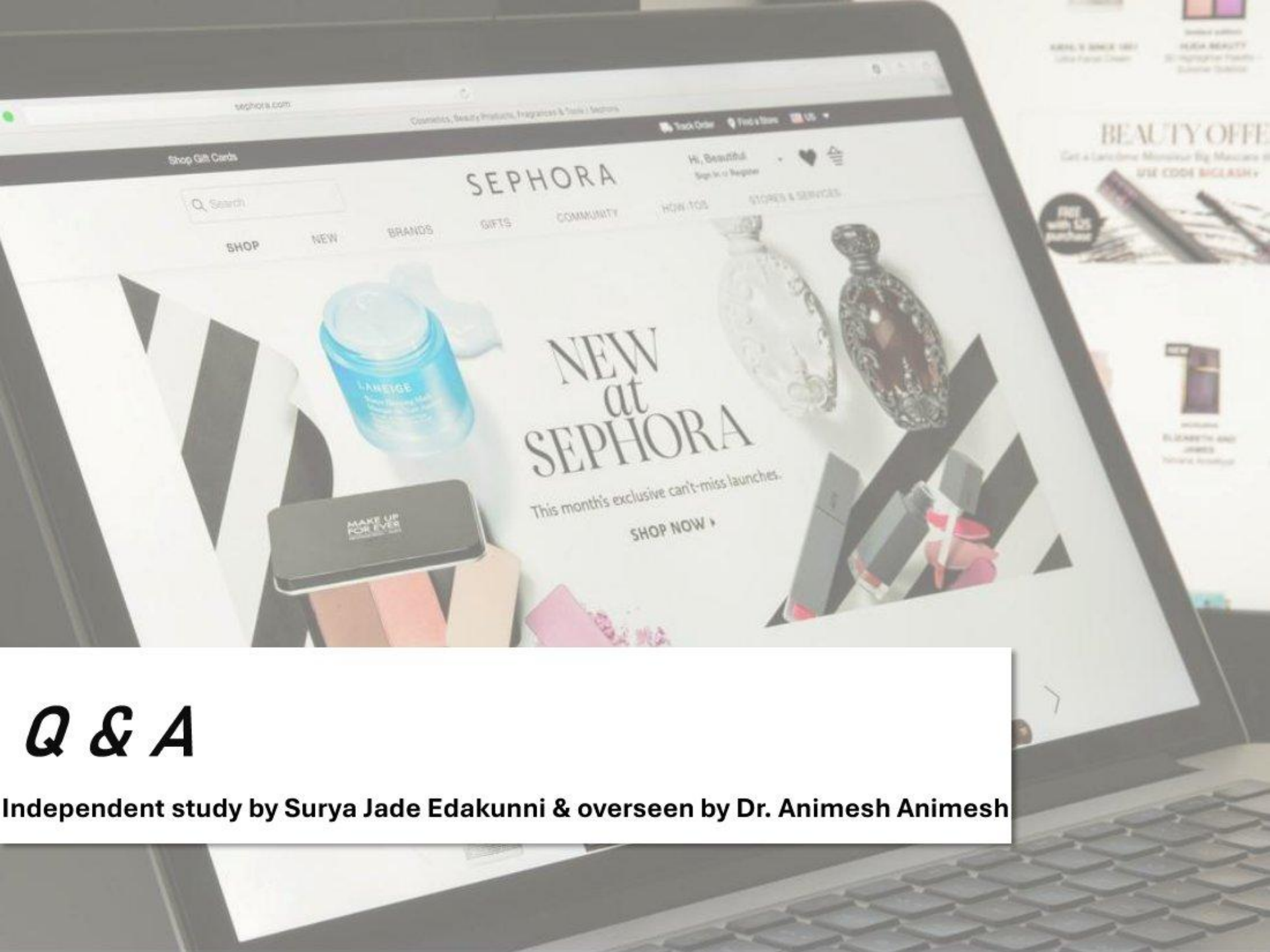


Multimodal Search
- Image-based product search
- Multimodal text + image embeddings
- Refine visual searches using text requirements

LLM Improvements
- Systematically compare models on the same product queries
- Test whether requirements and retrieved evidence are preserved
- Explore stronger prompting, model selection or retail-specific fine-tuning

Personalization
- Incorporate previous purchases, searches and preferences
- Use customer-product relationships as additional ranking evidence
- Combine semantic relevance with customer behaviour

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Q & A

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